

Solutions for Algebraic Geometry Hartshone

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1 Varieties

2 Schemes

2.1 Sheaves

Question 2.1: Flasque sheaf

- (1) A constant sheaf on an irreducible topological space is flabby (flasque).
 (2) Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be a short exact sequence of sheaves with $\mathcal{F}'$ being flabby. For any open set U (for any local section functor $\Gamma(U, -)$), the sequence of abelian groups

$$0 \rightarrow \Gamma(U, \mathcal{F}') \rightarrow \Gamma(U, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F}'') \rightarrow 0$$

is exact.

- (3) Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be a short exact sequence of sheaves. If $\mathcal{F}', \mathcal{F}$ are flabby, then $\mathcal{F}''$ is. If $\mathcal{F}'$ and $\mathcal{F}''$ are flabby, then $\mathcal{F}$ is.
 (4) Let $f : X \rightarrow Y$ be a continuous map. If $\mathcal{F}$ is a flabby sheaf on X , then $f_*\mathcal{F}$ is a flabby sheaf on Y .
 (5)

Proof: (1) Let X be an irreducible topological space. Take an inclusion $U \hookrightarrow V$, with $U, V \subseteq X$. U and V are connected in an irreducible topological space. So locally constant functions of U and V are globally constant¹. The map $\Gamma(V, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{F})$ is surjective. Because for any $s \in \Gamma(U, \mathcal{F})$, s is a constant function on U : $\forall x \in U$, $s(x) = a$ for some $a \in A$ in the abelian group A . Consider a function $\tilde{s}(y) = a$ for every $y \in V$. Indeed, this function $\tilde{s} \in \Gamma(V, \mathcal{F})$. The projection of this $\tilde{s} \mapsto \tilde{s}|_U$ is exactly s . Hence, this map is surjective and $\mathcal{F}$ is flabby.

(2) Pick an open cover $\{U_i\}$ of U . The exactness of $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ is equivalent to the exactness of abelian groups

$$0 \rightarrow \Gamma(U_i, \mathcal{F}') \xrightarrow{\alpha_{U_i}} \Gamma(U_i, \mathcal{F}) \xrightarrow{\beta_{U_i}} \Gamma(U_i, \mathcal{F}'') \rightarrow 0$$

for each U_i . Pick an element $s \in \mathcal{F}''(U)$. Denote each restriction of s on U_i to be $s_i := s|_{U_i}$. Clearly, $s_i \in \Gamma(U_i, \mathcal{F}'')$. Using the surjectivity of β_{U_i} , we can find $t_i \in \Gamma(U_i, \mathcal{F})$ such that $\beta_{U_i}(t_i) = s_i$.

By Zorn's lemma, there exists a maximal $V \subseteq U$ among all possible open sets admitting a lifted section, and an element $t_V \in \Gamma(V, \mathcal{F})$ such that $\beta_V(t_V) = s|_V$.

(i) If $V = U$, done.

(ii) If $V \neq U$, then pick an $x \in U \setminus V$ and a neighborhood W of x . V must intersect W . Otherwise, this function t_V can be trivially extended to $V \cup W$, which violates the maximality of V . Since $\beta_{V \cap W}(t_V) = s|_{V \cap W} = \beta_{V \cap W}(t_W)$, we have $t_V - t_W \in \ker \beta_{V \cap W} = \text{im } \alpha_{V \cap W}$. So, $\exists r \in \Gamma(V \cap W, \mathcal{F}')$, such that

$$t_V - t_W = \alpha_{V \cap W}(r)$$

The property that $\mathcal{F}'$ is flabby enlightens us that this r can be extended from $V \cap W$ to the whole W . Namely, to an $\tilde{r} \in \Gamma(W, \mathcal{F}')$. Now, define a new element t_W^{new} to be

$$t_W^{\text{new}} := t_W + \alpha_W(\tilde{r}) \quad 2$$

Notice that $\beta_W(t_W^{\text{new}}) = \beta_W(t_W) + 0 = s|_W$, and $t_W^{\text{new}}|_{V \cap W} = t_V|_{V \cap W}$, because

$$t_W^{\text{new}}|_{V \cap W} = (t_W + \alpha_W(\tilde{r}))|_{V \cap W} = t_W|_{V \cap W} + \alpha_{V \cap W}(r) = t_W|_{V \cap W} + (t_V - t_W)|_{V \cap W} = t_V|_{V \cap W}.$$

Since $t_W^{\text{new}}|_{V \cap W} = t_V|_{V \cap W}$, the gluing axioms of a sheaf allow us to perfectly patch t_V and t_W^{new} to extend the section onto a larger space $V \cup W$, contradicting the maximality of V . So the only possibility is $V = U$.

¹Being locally constant on U means that for any $x \in U$ and an open neighborhood $W \ni x$, the function s is a constant function on W . Now let $\{U_i\}_i$ be an open cover of U , with s being constant on each U_i . So s has a value on U_i and has a value on U_j . If U_i intersects U_j , then s must have the same values on U_i and U_j . By the gluing property of $\mathcal{F}|_U$, s can be extended to a constant on U .

²The geometric intuition behind this construction is profound: we first measure the discrepancy between t_V and t_W on the overlap $V \cap W$, which can be visualized as a 'height difference' captured by the function r . We want to translate t_W to a new height so that this difference becomes zero, allowing them to agree on the intersection. Because $\mathcal{F}'$ is flabby, this local 'height adjustment' r can be extended to a global adjustment $\tilde{r}$ over the entirety of W . Adding this adjustment lifts t_W to t_W^{new} , resolving the mismatch on the boundary while preserving its projection to s .

(3) Let $U \subseteq V$ be open subsets of X . We need to show that the restriction map $\rho''_{V,U} : \mathcal{F}''(V) \rightarrow \mathcal{F}''(U)$ is surjective. Let $s \in \mathcal{F}''(U)$. Consider the following commutative diagram (since the morphisms in $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ are natural transformations) with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}'(V) & \xrightarrow{\alpha_V} & \mathcal{F}(V) & \xrightarrow{\beta_V} & \mathcal{F}''(V) \longrightarrow 0 \\ & & \downarrow \rho'_{V,U} & & \downarrow \rho_{V,U} & & \downarrow \rho''_{V,U} \\ 0 & \longrightarrow & \mathcal{F}'(U) & \xrightarrow{\alpha_U} & \mathcal{F}(U) & \xrightarrow{\beta_U} & \mathcal{F}''(U) \longrightarrow 0 \end{array} \quad \begin{array}{l} (\dagger) \\ (\ddagger) \end{array}$$

Because $\mathcal{F}'$ is flabby, $(\dagger)$ and $(\ddagger)$ are exact sequences (from part (2)) and $\rho'_{V,U}$ is surjective. $\mathcal{F}$ is flabby so $\rho_{V,U}$ is surjective.

For the s we fixed, using the surjectivity of β_U and $\rho_{V,U}$, there exists an $\tilde{s} \in \mathcal{F}(V)$ such that $(\beta_U \circ \rho_{V,U})(\tilde{s}) = s$. Then $\rho''_{V,U}$ is surjective because for $t := \beta_V(\tilde{s})$, we have $\rho''_{V,U}(t) = \rho''_{V,U}(\beta_V(\tilde{s})) = \beta_U(\rho_{V,U}(\tilde{s})) = s$.

Now let $\mathcal{F}'$ and $\mathcal{F}''$ be flabby. We want to show that $\rho_{V,U} : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ is surjective. Still, consider a similar commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{F}'(V) & \xrightarrow{\alpha_V} & \mathcal{F}(V) & \xrightarrow{\beta_V} & \mathcal{F}''(V) \longrightarrow 0 \\ & & \downarrow \rho'_{V,U} & & \downarrow \rho_{V,U} & & \downarrow \rho''_{V,U} \\ 0 & \longrightarrow & \mathcal{F}'(U) & \xrightarrow{\alpha_U} & \mathcal{F}(U) & \xrightarrow{\beta_U} & \mathcal{F}''(U) \longrightarrow 0 \end{array} \quad \begin{array}{l} (\dagger) \\ (\ddagger) \end{array}$$

For the same reasons, $(\dagger)$ and $(\ddagger)$ are exact sequences (from part (2)) and $\rho'_{V,U}$ is surjective. $\mathcal{F}''$ is flabby so $\rho''_{V,U}$ is surjective.

Pick an element $s \in \mathcal{F}(U)$, sending it via β_U to $t = \beta_U(s) \in \mathcal{F}''(U)$. By the surjectivity of $\rho''_{V,U}$ and β_V , $\exists \tilde{s} \in \mathcal{F}(V)$ such that $\rho''_{V,U}(\beta_V(\tilde{s})) = t$. Restrict this $\tilde{s}$ via $\rho_{V,U}$ and compare the difference $\rho_{V,U}(\tilde{s}) - s$. Projecting this difference via β_U yields zero, meaning it lies in $\ker \beta_U = \text{im } \alpha_U$. So, $\exists r \in \mathcal{F}'(U)$ such that $\rho_{V,U}(\tilde{s}) - s = \alpha_U(r)$. The surjectivity of $\rho'_{V,U}$ lifts r back to $\tilde{r} \in \mathcal{F}'(V)$ such that $\rho'_{V,U}(\tilde{r}) = r$. So far, we have

$$(\alpha_U \circ \rho'_{V,U})(\tilde{r}) = \rho_{V,U}(\tilde{s}) - s$$

By the commutativity of the leftmost square,

$$(\rho_{V,U} \circ \alpha_V)(\tilde{r}) = \rho_{V,U}(\tilde{s}) - s$$

which implies that $\rho_{V,U}(\tilde{s} - \alpha_V(\tilde{r})) = s$. Thereby, the preimage of s is $\tilde{s} - \alpha_V(\tilde{r})$, proving that $\rho_{V,U}$ is surjective. $\square$

2.2 Schemes

Question 2.2

Let A be a ring and $X = \text{Spec} A$, let $f \in A$ and $D(f) := X \setminus V(\langle f \rangle)$. Show that

$$(D(f), \mathcal{O}_X|_{D(f)}) \cong (\text{Spec} A_f, \mathcal{O}_{\text{Spec} A_f})$$

Question 2.3

Let $(X, \mathcal{O}_X)$ be a scheme and $U \subseteq X$ be an open subset. Show that the *induced scheme* on U , $(U, \mathcal{O}_X|_U)$ is a scheme.

Question 2.4

- (1) Show that $(X, \mathcal{O}_X)$ is reduced $\Leftrightarrow \forall P \in X$, the local ring $\mathcal{O}_{X,P}$ has no nilpotent element.
- (2) Let $(X, \mathcal{O}_X)$ be a scheme and $(\mathcal{O}_X)_{\text{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$, where for any ring A , $A_{\text{red}} := A/I$ with I the ideal of nilpotent elements of A . Show that the *reduced scheme* $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme and there is a morphism of schemes $(X, (\mathcal{O}_X)_{\text{red}}) \rightarrow (X, \mathcal{O}_X)$, which is a homeomorphism on the underlying topological space.
- (3)

Question 2.5

Given a morphism $f : X \rightarrow \text{Spec} A$, and a associated map on sheaves $f^\# : \mathcal{O}_{\text{Spec} A} \rightarrow f_* \mathcal{O}_X$, we obtain a homomorphism $A \rightarrow \Gamma(X, \mathcal{O}_X)$ by taking global sections.

Thus, there is a natural map

$$\alpha : \text{Mor}_{\mathbf{Sch}}((X, \mathcal{O}_X), (\text{Spec } A, \mathcal{O}_{\text{Spec } A})) \rightarrow \text{Mor}_{\mathbf{CRing}}(A, \Gamma(X, \mathcal{O}_X)) \quad a$$

Show that α is a bijection map.

^aHom is synonymous to Mor.

Proof:

□

Question 2.6: Final objects in category of schemes

Describe $\text{Spec } \mathbb{Z}$. Show that $\text{Spec } \mathbb{Z}$ is a final object for the category of schemes.

Proof: Let $(X, \mathcal{O}_X)$ be an arbitrary scheme. By definition of schemes, $\exists$ an open cover of X , $\{U_i\}_{i \in I}$, such that $\forall i$, $(U_i, \mathcal{O}_X|_{U_i})$ is an affine scheme. i.e. $\forall i$, $\exists R_i$ a ring, such that $(U_i, \mathcal{O}_X|_{U_i}) \cong (\text{Spec } R_i, \mathcal{O}_{\text{Spec } R_i})$.

Since $\mathbb{Z}$ is initial in the category $\mathbf{CRing}$, then $\forall i$, $\exists!$ ring homomorphism $\mathbb{Z} \rightarrow R_i$ which induces unique morphism of affine schemes $(\text{Spec } R_i, \mathcal{O}_{\text{Spec } R_i}) \rightarrow (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}})$. Hence, there is a unique morphism $(U_i, \mathcal{O}_X|_{U_i}) \rightarrow (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}})$ by composing the above-mentioned isomorphism.

This result lifts to the unique morphism of schemes $(X, \mathcal{O}_X) \rightarrow (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}})$ because the Hom of scheme, $H := \text{Hom}_{\mathbf{Sch}}((X, \mathcal{O}_X), (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}}))$ is a sheaf.

$$H(U) := \text{Hom}_{\mathbf{Sch}}((U_i, \mathcal{O}_X|_{U_i}), (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}}))$$

(So, the locality and gluing axioms extend morphisms on $(U_i, \mathcal{O}_X|_{U_i})$ globally.)

□

Question 2.7: Initial objects in category of schemes

Describe $\text{Spec } \{0\}$. Show that $\text{Spec } \{0\}$ is an initial object for the category of schemes.

Question 2.8: Characterisation of tangent spaces

Let $(X, \mathcal{O}_X)$ be a scheme over a field k and $k[\epsilon]/\langle \epsilon^2 \rangle$ be the ring of dual numbers over k .

Show that giving a k -morphism $\varphi : \text{Spec } k[\epsilon]/\langle \epsilon^2 \rangle \rightarrow X \Leftrightarrow$ giving $x \in X$, rational over k and an element of Zariski tangent space T_x .

Remark Some analysis on $k[\epsilon]/\langle \epsilon^2 \rangle$:

(1) When $k = \mathbb{R}$, $\forall f(x) = \sum_i a_i x^i \in (\mathbb{R}[\epsilon]/\langle \epsilon^2 \rangle)[x]$, $f(\overline{a + b\epsilon}) = f(\bar{a}) + \bar{b}f'(\bar{a})\bar{\epsilon}$.

(2) For general fields k , units are elements of the form $a + \langle \epsilon^2 \rangle$ or $a + b\epsilon + \langle \epsilon^2 \rangle$ with $a, b \neq 0$. So, non-units of $k[\epsilon]/\langle \epsilon^2 \rangle$ are of the form $b\epsilon + \langle \epsilon^2 \rangle$ with $b \neq 0$, which are in $\langle \epsilon \rangle/\langle \epsilon^2 \rangle$.

Since every proper ideal cannot have any units, so it is a subset of the set of non-units. Hence a subset of $\langle \epsilon \rangle/\langle \epsilon^2 \rangle$. Therefore, $\langle \epsilon \rangle/\langle \epsilon^2 \rangle$ is the unique maximal ideal of $k[\epsilon]/\langle \epsilon^2 \rangle$. Thus, $k[\epsilon]/\langle \epsilon^2 \rangle$ is local.

Proof: $\Rightarrow$ Consider we have a morphism $\varphi : \text{Spec } k[\epsilon]/\langle \epsilon^2 \rangle \rightarrow X$. Then, x can be given by the point that the unique maximal ideal of $k[\epsilon]/\langle \epsilon^2 \rangle$ goes, i.e. $x := \varphi(\langle \epsilon \rangle/\langle \epsilon^2 \rangle)$

□

Question 2.9: Anti-equivalence of Fld and Sch

The category of field **Fld** is antiequivalent to the category of scheme **Sch**.

Question 2.10: Generic points

Question 2.11: Spec $\mathbb{R}[x]$ **Question 2.12: Spec $\mathbb{F}_p[x]$** **Question 2.13: Glueing lemma****Question 2.14: Quasi-compactness of a topological space**

- (1) Show that a topological space is Noetherian $\Leftrightarrow$ every *open subset* is quasi-compact.
- (2) Let $(X, \mathcal{O}_X)$ be an affine scheme. Show that X is quasi-compact but not necessarily Noetherian.
- (3) If A is a Noetherian ring, then $\text{Spec}(A)$ is a Noetherian topological space.
- (4) The converse of (3) is not always true. Give an example (A is not Noetherian ring but $\text{Spec}(A)$ is a Noetherian topological space).

Question 2.15: Projective scheme

- (1) Let S be a graded ring. Show that $\text{Proj } S = \emptyset$ iff $\forall x \in S_+, x$ is nilpotent.
- (2) Let $\varphi : S \rightarrow T$ be a graded homomorphism of graded rings. $U := \{[\mathfrak{p}] \in \text{Proj } T \mid \varphi(S_+) \not\subseteq \mathfrak{p}\}$. Show that U is an open subset of $\text{Proj } T$ and φ determines a natural morphism $f : U \rightarrow \text{Proj } S$.
- (3) The morphism f can be an isomorphism even when φ is not an isomorphism. Show the following example:
Let d_0 be an integer. $\forall d \geq d_0, \varphi_d : S_d \rightarrow T_d$ is an isomorphism, then $U = \text{Proj } T$ and the morphism $f : \text{Proj } T \rightarrow \text{Proj } S$ is isomorphic.
- (4) Let V be a projective variety with homogeneous coordinate ring S . Show that $t(V) \cong \text{Proj } S$.

Proof: (1) $\Rightarrow$ Suppose $\exists y \in S_+$ that is not a nilpotent. Then, we will show that there is a homogeneous prime ideal that satisfies $S_+ \not\subseteq$ this ideal.

$y = \sum_{i_1 \leq d \leq i_m} y_d$ with each $y_d \in S_d$. Then, y is not a nilpotent implies that at least one y_t in the sum is not a nilpotent. Let $T := \{1, y_t, y_t^2, \dots\}$. Consider the localization of $T^{-1}S =: S_{y_t}$. Since y_t is not a nilpotent, $0 \notin T$ and S_{y_t} is not a zero ring, which implies that the degree-0 localization $S_{(y_t)}$ is not a zero ring.³ Furthermore, $\text{Spec } S_{(y_t)}$ is non-empty because every non-zero ring has at least one maximal ideal. The following isomorphism

$$D_+(y_t) \cong \text{Spec } S_{(y_t)}$$

implies that $D_+(y_t) \neq \emptyset \Rightarrow \text{Proj } S \neq \emptyset$. ($D_+(y_t) \subseteq \text{Proj } S$)

$\Leftarrow$ Suppose that $\forall x \in S_+, x$ is nilpotent. We will show that there is no homogeneous prime ideal $\mathfrak{p}$ that $S_+ \not\subseteq \mathfrak{p}$. Let $\mathfrak{p}$ be a homogeneous prime ideal. $\forall x \in S_+$, since $\exists n, x^n = 0 \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$ because $\mathfrak{p}$ is prime, implying $S_+ \subseteq \mathfrak{p}$ for all homogeneous prime ideal $\mathfrak{p}$.

(2) □

Question 2.16

- (1) Let V be a variety over the algebraically closed field k . Show that $P \in t(V)$ is a closed point $\Leftrightarrow$ the residue field of P is k .

³This is true for all S_y such that $y \in S_d$ with $d \geq 1$.

Question 2.17

Let $(X, \mathcal{O}_X)$ be a scheme and $f \in \Gamma(X, \mathcal{O}_X)$ be a global section. $\mathfrak{m}_x$ is the maximal ideal of the local ring $\mathcal{O}_{X,x}$. $X_f := \{x \in X : f_x \notin \mathfrak{m}_x\}$ where $f_x \in \mathcal{O}_{X,x}$ or a germ of the function f .

- (1) Suppose $(U, \mathcal{O}_X|_U)$ is an *open affine scheme* of $(X, \mathcal{O}_X)$ with $U = \text{Spec} B$. $f|_U \in \Gamma(U, \mathcal{O}_X|_U) \cong B$. Show that $X_f \cap U = D(f|_U)$. Conclude that X_f is an open subset of X .
- (2) Let $(X, \mathcal{O}_X)$ be quasi-affine and $a \in \Gamma(X, \mathcal{O}_X)$ such that $a|_{X_f} = 0$. Show that $\exists n > 0, f^n a = 0$.

Proof: (1) $U \cap X_f = \{x \in U | f_x \notin \mathfrak{m}_x\}$
(2) Let $\square$

Question 2.18: A criterion for affineness

Question 2.19: Properties of a ring homomorphism to the induced morphisms

- (1) Let A be a ring, $X = \text{Spec} A$ and $f \in A$. f is nilpotent $\Leftrightarrow D(f) = \emptyset$.
- (2) Let $\varphi : A \rightarrow B$ be a homomorphism of rings and let $f : Y = \text{Spec} B \rightarrow \text{Spec} A = X$ be the induced morphism of affine schemes. φ is injective $\Leftrightarrow f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$.
- (3) φ is surjective $\Leftrightarrow f : Y \rightarrow C^a$ is a homeomorphism and $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is surjective. Hence,

^a C is a closed subset of X

Proof: (1) Suppose that f is a nilpotent, then $\exists n$, such that $f^n = 0$. This means that $\forall \mathfrak{p} \in \text{Spec} A, f^n \in \mathfrak{p} \Rightarrow f \in \mathfrak{p}$.

$$D(f) = \{[\mathfrak{p}] \in \text{Spec} A : f^n \notin \mathfrak{p}\} = \emptyset$$

$D(f) = \emptyset \Leftrightarrow \forall$ prime ideal $\mathfrak{p}, f \in \mathfrak{p}$. $\{0\}$ is also a prime ideal. Thereby, $f = 0$ and f is a nilpotent.

(2) $\varphi : A \rightarrow B$ is injective if and only if $\forall \mathfrak{p}, \varphi : A_{\mathfrak{p}} \rightarrow B \otimes_A A_{\mathfrak{p}}$ is injective.

In the meantime, $f^\#$ is injective if and only if $f_{\mathfrak{p}}^\# : (\mathcal{O}_X)_{\mathfrak{p}} \rightarrow (f_* \mathcal{O}_Y)_{\mathfrak{p}}$ is injective.

So the only thing is to see that $(\mathcal{O}_X)_{\mathfrak{p}} \cong A_{\mathfrak{p}}$ and $(f_* \mathcal{O}_Y)_{\mathfrak{p}} \cong B \otimes_A A_{\mathfrak{p}}$. The first one is clear and the second one is because

$$\begin{aligned} (f_* \mathcal{O}_Y)_{\mathfrak{p}} &= \lim_{\rightarrow \mathfrak{p} \in D(g)} f_* \mathcal{O}_Y(D(g)) \\ &= \lim_{\mathfrak{p} \in D(g)} \mathcal{O}_Y(f^{-1}(D(g))) \\ &= \lim_{\mathfrak{p} \in D(g)} \mathcal{O}_Y(D(\varphi(g))) \\ &= \lim_{\mathfrak{p} \in D(g)} B_{\varphi(g)} & \mathcal{O}_Y(D(\varphi(g))) \cong B_g \\ &\stackrel{\dagger}{=} (\varphi(A \setminus \mathfrak{p}))^{-1} B \cong B \otimes_A A_{\mathfrak{p}} \end{aligned}$$

$\square$

Question 2.20

Let A be a ring commutative?. Show TFAE:

- (1) $\text{Spec } A$ is disconnected.
- (2) $\exists$ non-zero orthogonal idempotents $e_1, e_2 \in A$, i.e. $e_1^2 = e_1, e_2^2 = e_2, e_1 e_2 = 0$ and $e_1 + e_2 = 1$.
- (3) $A \cong A_1 \times A_2$, where A_1, A_2 are two non-zero rings.

Proof: (1) $\Rightarrow$ (2)

(2) $\Rightarrow$ (1) Consider the distinguishable open set of e_1 , $D(e_1)$. To see $\text{Spec} A$ is disconnected, we want to show that $D(e_1)$ is a non-trivial clopen set.

For the clopenness, $D(e_1)$ itself is already open. Also, $D(e_1) = \text{Spec} A - D(e_2)$. So $D(e_1)$ is closed.

(2) $\Rightarrow$ (3) Let $A_1 := \langle e_1 \rangle$ and $A_2 := \langle e_2 \rangle$. Then, A_1 and A_2 are rings with multiplicative identities e_1 and e_2

respectively ⁴. The homomorphism is defined to be

$$\varphi : A_1 \times A_2 \rightarrow A \quad (ae_1, be_2) \mapsto ae_1 + be_2$$

This is indeed a ring homomorphism. It is surjective because for any $r \in A$, $r = r(e_1 + e_2)$ is the image of (re_1, re_2) . To show this is injective, for any element $(ae_1, be_2) \in \ker \varphi$. $ae_1 + be_2 = 0$. From $ae_1 = -be_2$, $a = (a - b)e_2$ and $b = (b - a)e_1$. So,

$$(ae_1, be_2) = ((a - b)e_2e_1, (b - a)e_1e_2) = (0, 0)$$

(3) $\Rightarrow$ (2) Suppose that $A_1 \times A_2 \cong A$. Let φ be the isomorphism $\varphi : A_1 \times A_2 \rightarrow A$. Let

$$e_1 := \varphi(1_{A_1}, 0) \quad e_2 := \varphi(0, 1_{A_2})$$

e_1 and e_2 are orthogonal idempotents. □

2.3 First properties of schemes

3 Cohomology

4 Curves

5 Surfaces

⁴There's a subtlety that A_1 and A_2 are rings. We didn't say that they are subrings of A .